

## Snedecor's F-Distribution

A random variable  $F$  is said to follow snedecor's  $F$ -distribution or simply  $F$ -distribution if its probability density function is given by

$$f(F) = \frac{(v_1/v_2)^{v_1/2}}{\beta\left(\frac{v_1}{2}, \frac{v_2}{2}\right)} \cdot \frac{F^{v_1/2-1}}{\left(1 + \frac{v_1 F}{v_2}\right)^{(v_1+v_2)/2}}, \quad F > 0.$$

## Properties of the F-Distribution

1. The probability curve of the  $F$ -distribution is roughly sketched in Fig. 8.3.

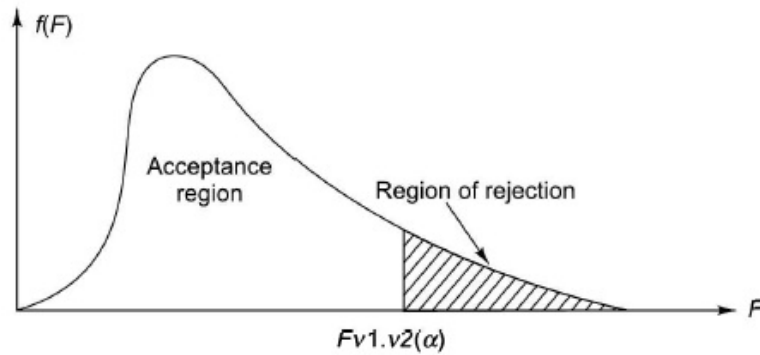


Fig. 8.3

2. The square of the  $t$ -variate with  $n$  degrees of freedom follows a  $F$ -distribution with 1 and  $n$  degrees of freedom.
3. The mean of the  $F$ -distribution is  $\frac{v_2}{v_2 - 2}$  ( $v_2 > 2$ ).
4. The variance of the  $F$ -distribution is

$$\frac{2v_2^2(v_1 + v_2 - 2)}{v_1(v_2 - 2)^2(v_2 - 4)} \quad (v_2 > 4)$$

**F-test for Equality of Population Variances.** Suppose we want to test (i) whether two independent samples  $x_i$ , ( $i = 1, 2, \dots, n_1$ ) and  $y_j$ , ( $j = 1, 2, \dots, n_2$ ) have been drawn from the normal populations with the same variance  $\sigma^2$ , (say), or (ii) whether the two independent estimates of the population variance are homogeneous or not.

Under the null hypothesis ( $H_0$ ) that (i)  $\sigma_X^2 = \sigma_Y^2 = \sigma^2$ , i.e., *the population variances are equal* or (ii) *Two independent estimates of the population variance are homogeneous*, the statistic  $F$  is given by

$$F = \frac{S_X^2}{S_Y^2}$$

$$\text{where } S_X^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (x_i - \bar{x})^2 \text{ and } S_Y^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (y_j - \bar{y})^2$$

are unbiased estimates of the common population variance  $\sigma^2$  obtained from two independent samples and it follows Snedecor's  $F$ -distribution with  $(v_1, v_2)$  d.f. [where  $v_1 = n_1 - 1$  and  $v_2 = n_2 - 1$ ].

Two independent samples of 8 and 7 items respectively had the following values of the variable:

|           |    |    |    |    |    |   |    |    |
|-----------|----|----|----|----|----|---|----|----|
| Sample I  | 9  | 11 | 13 | 11 | 15 | 9 | 12 | 14 |
| Sample II | 10 | 12 | 10 | 14 | 9  | 8 | 10 |    |

Does the estimate of Population variance differ significantly? Given that for 7 degrees of freedom the value of F at 5 % level of significance is 4.20 nearly

Solution:-  
 Null Hypothesis ( $H_0$ ) :- Population variance does not differ significantly  
 Alternative Hypothesis ( $H_1$ ) :- Population variance differ significantly  
 Hypothesis represents a two tailed test

|               |                   |               |                  |
|---------------|-------------------|---------------|------------------|
| $x$           | $x^2$             | $y$           | $y^2$            |
| 9             | 81                | 10            | 100              |
| 11            | 121               | 12            | 144              |
| 13            | 169               | 10            | 100              |
| 11            | 121               | 14            | 196              |
| 15            | 225               | 9             | 81               |
| 9             | 81                | 8             | 64               |
| 12            | 144               | 10            | 100              |
| 14            | 196               |               |                  |
| $\sum x = 94$ | $\sum x^2 = 1138$ | $\sum y = 73$ | $\sum y^2 = 785$ |

Mean  $\bar{x} = \frac{94}{8}$

$\bar{y} = \frac{73}{7}$

Now

$$\begin{aligned} \sum_{i=1}^8 (x - \bar{x})^2 &= \sum_{i=1}^8 (x^2 - 2x\bar{x} + \bar{x}^2) \\ &= \sum_{i=1}^8 x^2 - 2\bar{x} \sum_{i=1}^8 x + \bar{x}^2 \sum_{i=1}^8 1 \\ &= 1138 - \left[ 2 \left( \frac{94}{8} \right) \times 94 \right] + \left[ \left( \frac{94}{8} \right)^2 \times 8 \right] = 33.5 \end{aligned}$$

$$\begin{aligned} \text{Similarly } \sum_{i=1}^7 (y - \bar{y})^2 &= \sum_{i=1}^7 y^2 - 2\bar{y} \sum_{i=1}^7 y + \bar{y}^2 \sum_{i=1}^7 1 \\ &= 785 - \left( 2 \times \frac{73}{7} \times 73 \right) + \left[ \left( \frac{73}{7} \right)^2 \times 7 \right] = 23.7 \end{aligned}$$

Hence  $S_x^2 = \frac{33.5}{7}$  ;  $S_y^2 = \frac{23.7}{6}$

Test statistic  $F = \frac{S_x^2}{S_y^2} = \frac{33.5 \times 6}{23.7 \times 7} = 1.21$

Since  $|F| < F_{0.05}$ , we accept the null Hypothesis.

In one sample of 8 observations, the sum of the squares of deviations of the sample values from the sample mean was 84.4 and in the other sample of 10 observations it was 102.6. Test whether this difference is significant at 5 per cent level, given that the 5 per cent point of  $F$  for  $n_1 = 7$  and  $n_2 = 9$  degrees of freedom is 3.29.

Soln - Given  $n_1 = 8$  ;  $n_2 = 10$  and

$$\sum_1^8 (x - \bar{x})^2 = 84.4 \quad ; \quad \sum_1^{10} (y - \bar{y})^2 = 102.6$$

Null Hypothesis: ( $H_0$ ) The difference between population variance is not significant

Alternative Hypothesis ( $H_1$ ) : The difference between population variance is significant.

$$\text{Now } S_x^2 = \frac{1}{n_1 - 1} \sum_1^8 (x - \bar{x})^2 = \frac{1}{7} 84.4$$

$$S_y^2 = \frac{1}{n_2 - 1} \sum_1^{10} (y - \bar{y})^2 = \frac{1}{9} 102.6$$

$$\text{Now, } F = \frac{S_x^2}{S_y^2} = \frac{84.4 \times 9}{102.6 \times 7} = 1.057$$

Since  $|F| < F_{0.05}$ , we accept the null Hypothesis.